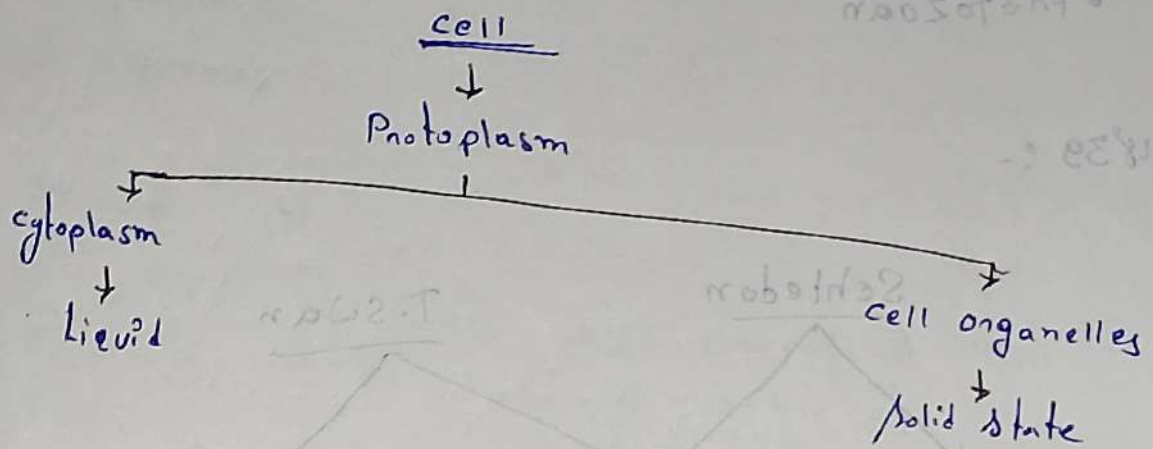


Cell

- Structural & functional unit



- Study of cell is known as cytology.
- Robert Hooke is "father of cell".
- The term cytology was given by Hertwig.
- Hertwig was the father of zoology.
- Father of modern cytology → Swan Son
- Father of cytogenetics → H.T. Mullen (mullen)
- Father of Indian cytology → A.C. SHARMA.

1665

↳ cell
↳ Robert Hook
↳ Bank of tree

1674 :- A.V. Leuwenhach

↳ Bank of tree
↓
Dead cell
↓
harden
↓
microscope.

- Living cell
- Bacteria.

Part of cell

- cell wall
- cell membrane
- cytoplasm
- Nucleus

cell wall

પૃથક 42 સારા જાણ amount જે પાત્ર
જાને ઘાસ carbohydrate જે Cellulose છે.

Cellulose + Hemicellulose

↓

Polysaccharide

→ 0% water

→ Hard

cell wall

Bacteria :- peptidoglycan + Amino acid + Muremic Acid.

Fungi → chitin.

Algae → cellulose + galactone + CaCO_3 (cell wall)

cell wall

↳ function

↳ shape, rigidity, protection.

• cell membrane :-

It is also known as

↳ plasma membrane

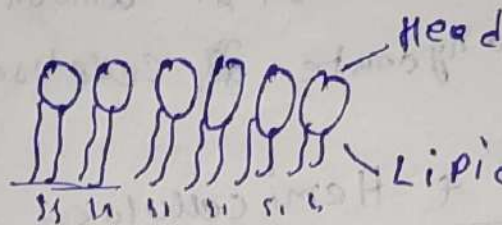
↳ Plasmalemma

↳ discovered by → ~~C. and Negi~~ C. and Negi

+

Singer + Nicolson :-

↳ "fluid mosaic model"



↳ Bi-layered

↳ Phospho-lipid

Thickness is $75 \text{ \AA}$

→ 7.5 nm

Donk :-

↳ Semipermeable membrane

(selective membrane)

★

NUCLEUS → Study karyology

↳ Discovered by Robert Brown

↳ Root cell of orchid plant

↳ Largest part of the cell

↳ Brain of cell / manager of cell.

Plant cell > Animal cell

Nucleus.

Mammals

↳ RBC → Brainless cell

↳ Nucleus.

Exception → (camel, llama) nucleus present inside RBC.

Phloem → sieve cells → nucleus

• Nucleus membrane

• Nucleoplasm

• Chromatin

• Nucleolus

▣ Nuclear membrane

→ Lipoprotein

→ for exchange of nutrients

▣ Nucleoplasm

↳ colourless

Chromatin

↳ Thread like structure

↳ nucleic acid

DNA → RNA

Nucleolus

↳ cell division.

Nucleic Acid → Friedrich Miescher

(Deoxy Ribose Nucleic Acid) DNA

RNA (Ribose Nucleic Acid)

DNA

↳ Watson & Crick

(Double helix Model)

↳ Mitochondria → DNA

↳ Chloroplast → DNA

Nucleotide

Nucleic Acid

DNA → RNA

Nucleotide = Nucleoside + Phosphoric acid

↓
Nitrogenous Base

↓
Sugar

DNA

2 - Pentose Sugar.

Histone protein.

Helix of
sugar-phosphates



cytosine | C

guanine | G

ADENINE | A

THYMINE | T

RNA

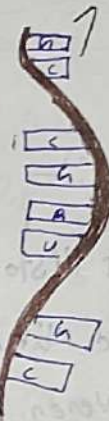
1 - Pentose Sugar

Histone protein. (x)

DNA

DEOXYRIBONUCLEIC Acid.

Nucleobases



cytosine | C

guanine | G

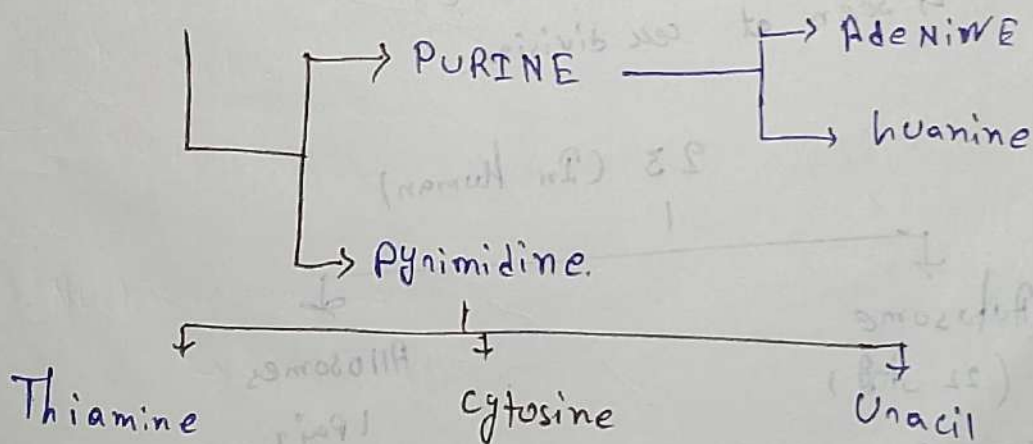
Adenine | A

Uracil | U

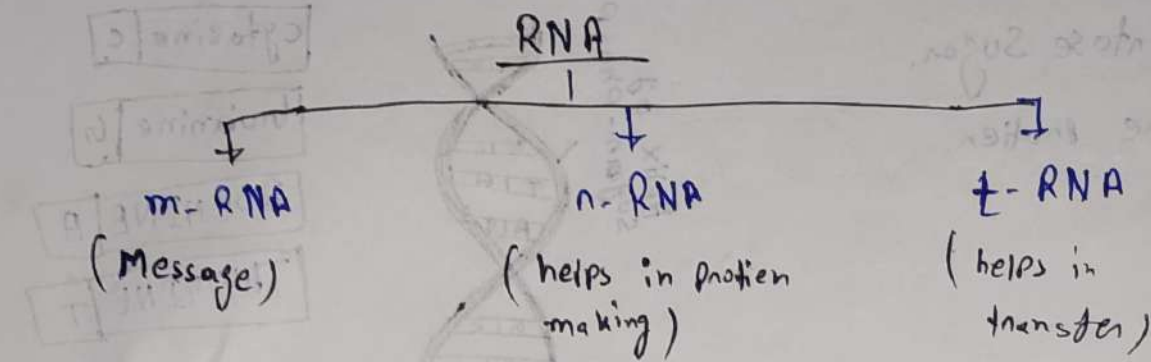
RNA

Ribonucleic Acid

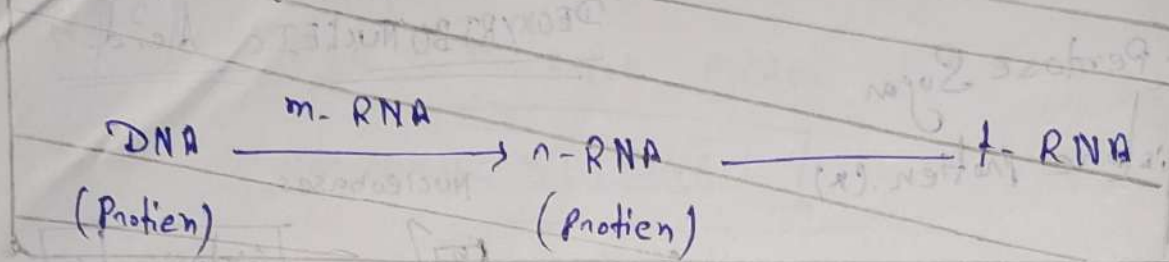
Nitrogenous Base



RNA :- helps in protein making.



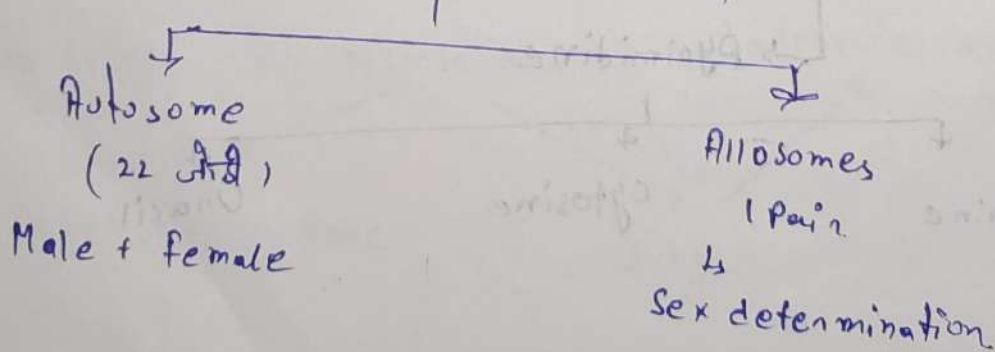
Translation :-



Chromosomes :-

- DNA + Histone Protein
- Thread like Structure
- discoverer was Punnett + Flemming
- Named by Waldeyer.
- 46 in human
- seen at cell division.

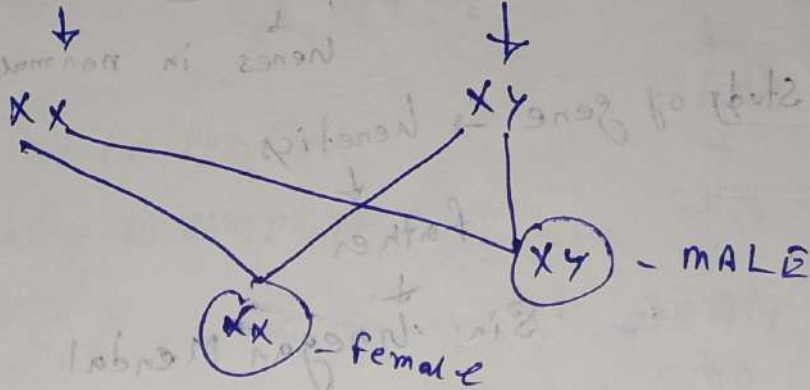
23 (In Human)



Sex chromosome

↓

OVA + Sperm



No. of chromosome in different species

- Human - 46
- Pea - 14
- Maize - 20
- monkey - 42
- Pigeon - 80
- Gold fish - 100
- chimpanzee - 48
- wheat - 42
- cat - 38
- elephant - 56
- ~~ant~~ ~~ant~~ ~~ant~~ ~~ant~~ - 1264 (632) Pair